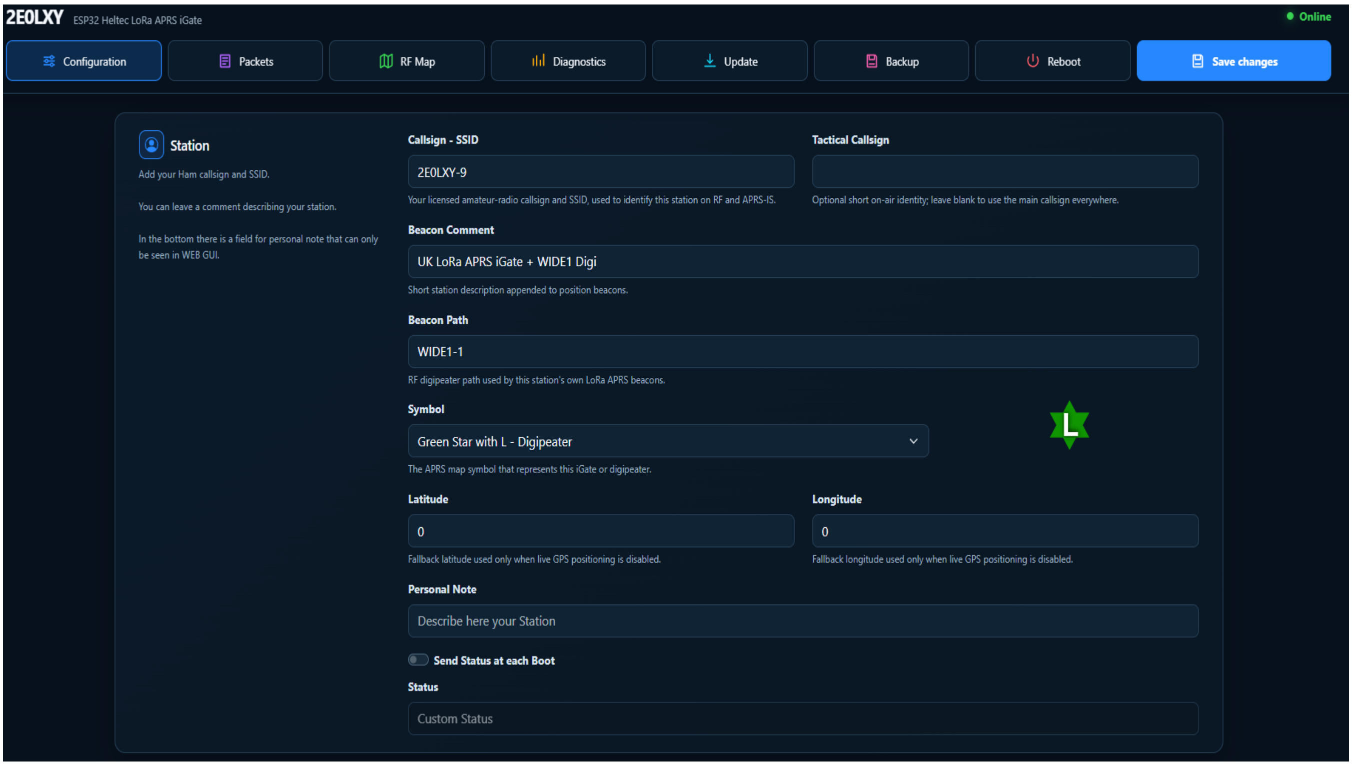
2E0LXY

LoRa APRS iGate and Digipeater

Installation, configuration and operation manual

Heltec WiFi LoRa 32 V3.2 and LilyGO LoRa32 V2.1

UK 439.9125 MHz profile - Firmware v1.1.6 - July 2026



1. Purpose

This firmware turns a Heltec WiFi LoRa 32 V3.2 or LilyGO TTGO LoRa32 V2.1 into a UK LoRa APRS receive iGate, APRS-IS connected station and configurable WIDE1/WIDE2 digipeater. It includes a modern browser interface, RF packet map, diagnostics, Wi-Fi scanning and board-safe GitHub OTA updates. The Heltec build also supports external GPS.

2. Safety and licence

- An amateur-radio licence and valid callsign are required for RF transmission.
- Check the current UK band plan and local coordination before transmitting.
- Use the minimum RF power that provides reliable local coverage.
- Back up configuration before updating firmware.
- Keep USB power connected throughout installation.
- The firmware is GPLv3 software supplied without warranty.

3. Hardware

Supported release target

Board	Processor	LoRa radio	Flash
Heltec WiFi LoRa 32 V3.2	ESP32-S3	SX1262	8 MB
LilyGO TTGO LoRa32 V2.1	ESP32	SX1278	4 MB

Each board has a separate OTA and full-flash image. Never cross-flash the images.

GPS wiring

GPS module	Heltec V3.2
TX	GPIO 47 (ESP32 RX)
RX	GPIO 48 (ESP32 TX)
GND	GND
Power	Use the voltage required by the GPS module

The firmware starts at 9600 baud and automatically checks 38400, 115200 and 4800 baud until it receives a checksum-valid NMEA sentence. Confirm the module voltage before applying power.

4. Installation

Browser installer

1. Open the 2E0LXY GitHub Pages installer in desktop Chrome or Edge.

2. Connect one supported board directly by USB.

3. Select **Connect and install**.

4. Choose its serial device. The installer detects ESP32-S3 or ESP32 and

selects the matching image.

5. Confirm installation and wait for verification and reboot.

The full installer can erase stored settings. Existing stations should download a configuration backup first.

Manual USB flash

Download the matching full-flash image from GitHub Releases and write it to address `0x0`:

- `2E0LXY-Heltec-V3.2-full-flash.bin` for Heltec ESP32-S3.
- `2E0LXY-LilyGO-LoRa32-V2.1-full-flash.bin` for LilyGO ESP32.

OTA update

Open **Update** in the device interface. The page compares the installed version with the latest 2E0LXY GitHub release. The compiled board identity selects the exact Heltec or LilyGO OTA filename. Installation is enabled only when that compatible asset is attached.

5. First connection

After boot, the iGate tries each saved Wi-Fi network. If none connects, its recovery access point becomes available. Connect to it and open `http://192.168.4.1/`. Change all default and recovery passwords before placing the device in service.

6. Recommended UK LoRa APRS profile

Parameter	Value
RX frequency	439.9125 MHz
TX frequency	439.9125 MHz
Spreading factor	SF12
Signal bandwidth	125 kHz
Coding rate	4/5
TX power	Use the minimum needed

Both ends of an RF link must use matching modulation parameters.

7. Station configuration

- **Callsign - SSID:** licensed callsign and station SSID.
- **Tactical callsign:** optional short identity.
- **Beacon comment:** concise public station description.
- **Beacon path:** path for this station's own RF beacon.
- **Symbol:** map symbol for an iGate or digipeater.
- **Latitude/longitude:** safe fallback while GPS is disabled, unavailable

or waiting for a valid fix.

- **GPS beacon:** automatically replaces the fallback with live coordinates

after a valid fix.

- **Position ambiguity:** reduces public coordinate precision.

The live **GPS Receiver** panel separately reports hardware/UART detection and position-fix validity. It shows satellites, HDOP, coordinates, altitude, speed, course, GPS UTC time, NMEA checksum totals and data age. Saving the GPS enable switch restarts the iGate so the UART can be changed safely.

For this third iGate, `2E0LXY-9` is used. Avoid SSID `-10` where it is already assigned to another station.

8. Wi-Fi

Select **Scan nearby Wi-Fi**, choose a network, enter its passphrase and save. Multiple networks may be stored. The startup delay allows a router or mobile hotspot time to become available after a power failure.

Never publish a configuration backup: it contains network and management credentials.

9. APRS-IS

Recommended settings:

Setting	Value
Server	`www.aprsnet.uk`
Port	`14580`
Filter	`m/100`
Passcode	Callsign-specific APRS-IS passcode

APRS Net UK provides a live APRS map, LoRa views, messaging, weather, watchlists, alerts and utilities. A free member account adds persistent watchlists, synced alerts and offline message storage.

Messages or objects should be gated from APRS-IS to RF only when useful locally and permitted by the routing safeguards.

10. Digipeater

- **Off:** no RF packet repeating.
- **WIDE1 fill-in:** local fill-in operation.
- **WIDE2 + WIDE1:** wider digipeater behaviour.
- **Backup digipeater:** falls back to local repeating if Wi-Fi or

APRS-IS is unavailable.

- **Eco mode:** reduces power consumption but can disable parts of the

display, networking or web interface.

Use a conservative beacon interval and path. Excessive repeating creates collisions and reduces channel capacity.

11. RF packets and map

The **Packets** page lists recently received RF packets with local NTP time, RSSI and SNR. The map's **Show** selector provides **RF only**, **APRS-IS** and **All** views. RF history remains separate so a busy internet feed cannot displace packets actually heard or transmitted by the radio. Only packets containing a supported APRS position are mapped.

12. Diagnostics

The diagnostics page reports:

- Valid RF packets.
- Duplicates suppressed.
- CRC failures and other radio errors.

- Successful transmissions and TX errors.
- Free heap and Wi-Fi RSSI.
- Heard stations with packet count, RSSI, SNR and frequency error.

CRC-failed frames are counted but never repaired, gated or repeated.

13. Time

The NTP section displays the actual device time, synchronization state, server and GMT correction. Use GMT offset `0` in UK winter and `1` during British Summer Time. Confirm the displayed time after changing the offset.

14. Backup, restore and security

- Download a configuration backup before major changes.
- Store backups privately because they contain passwords.
- Enable web authentication and use a unique password.
- Protect OTA credentials.
- Limit remote management to trusted callsigns and RF-only operation

where appropriate.

- Do not expose the HTTP interface directly to the public internet.

15. Firmware updates

The GitHub updater performs these checks:

1. Read the latest release from the standalone 2E0LXY repository.
2. Confirm that the matching Heltec or LilyGO OTA image is attached.
3. Compare semantic versions.
4. Require operator confirmation.
5. Download and write the firmware.
6. Verify the write and reboot.

Manual ElegantOTA upload remains available as a recovery route.

16. Tracker interoperability

The receiver is intended for standard LoRa APRS frames and Base91 APRS positions using matching radio settings. Battery telemetry, GPS positions and normal APRS paths are processed by the existing APRS packet pipeline.

Tracker SmartBeaconing, motion turn detection, tracker push-buttons and GPS power toggling are transmitter functions and do not belong in a fixed iGate. Experimental APRS434 18-byte address-compressed frames are not claimed as compatible until decoding and gating are validated with captured test vectors.

17. Troubleshooting

Web page is unavailable

Confirm the displayed IP address, Wi-Fi connection and eco mode. Ping the device, reconnect by USB and inspect serial output at 115200 baud.

APRS-IS will not connect

Check DNS, server, port, passcode and filter. Confirm the callsign is valid and the device has internet access.

No RF packets

Check antenna connection, RX enabled state, frequency and LoRa parameters. Compare diagnostics counters and verify that another station uses the same profile.

GPS does not fix

Confirm TX-to-RX crossover, GPIO 47/48, automatic baud scanning, antenna view of the sky and adequate module power. A cold start may take several minutes.

OTA fails

Keep power connected, verify internet access and retry manual firmware upload. Do not upload the full-flash image through the OTA form.

18. Project links

- Source: https://github.com/2E0LXY/2E0LXY-LoRa-APRS-iGate
- Releases: https://github.com/2E0LXY/2E0LXY-LoRa-APRS-iGate/releases
- Installer: https://2e0lxy.github.io/2E0LXY-LoRa-APRS-iGate/
- APRS Net UK: https://www.aprsnet.uk/

19. Acknowledgement

Thank you to CA2RXU for creating the original LoRa APRS iGate project. Original contributor notices remain in the source where required.